



Dictionary Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to OOPs.

Section 1: Dictionary Basics and Creation

★ Create and Display a Dictionary

★ Access Value using Key

★ Add a New Key-Value Pair

★ Update Value of an Existing Key

Access Value using get() Method

★ Find Length of a Dictionary

Nested Dictionary Creation and Access

Section 2: Adding, Removing, and Checking

★ Remove a Key-Value Pair using pop()

Remove a Key-Value Pair using del Statement

★ Check if a Key Exists in a Dictionary

Check if a Value Exists in a Dictionary

★ Clear All Elements from a Dictionary

Merge Two Dictionaries

Section 3: Iterating and Viewing

★ Iterate through Keys of a Dictionary

★ Iterate through Values of a Dictionary

★ Iterate through Key-Value Pairs using items()

Get All Keys as a List

Get All Values as a List

Section 4: Searching and Aggregation

★ Find Maximum Value in a Dictionary

★ Find Minimum Value in a Dictionary

Find Key with Maximum Value

★ Sum of All Values in a Dictionary

Sort Dictionary by Keys



★ Sort Dictionary by Values

Section 5: Dictionary Comprehension

★ Create a Dictionary using Dictionary Comprehension

★ Square of Numbers as Key-Value Pairs

Filter Dictionary based on Value Condition

Swap Keys and Values in a Dictionary

Invert a Dictionary with Duplicate Values (Group Keys)

Section 6: Dictionary-Based Logical Problems

★ Count Frequency of Words in a Sentence

★ Count Frequency of Characters in a String

Find Common Keys between Two Dictionaries

★ Merge Two Dictionaries Summing Values of Common Keys

Group List of Tuples into a Dictionary

Find Duplicate Values in a Dictionary

Convert Two Lists into a Dictionary (zip)

Section 7: Real World Dictionary Applications

★ Student Record Management using Dictionary

★ Employee Salary Lookup System

Inventory Management using Dictionary

★ Phone Book / Contact Directory using Dictionary

Voting/Poll Counter using Dictionary

★ Login Credential Validator using Dictionary